Lecture 10: Goodness-of-Fit Tests

SDS8102

Week 10

Overview. A model can match a sample mean and still miss the shape of the distribution. We test whether the whole distribution agrees with a proposed model. *The lecture asks two linked questions: which part of the data–model discrepancy should we measure, and how large can it be from sampling noise alone?*

We recall Week 6's empirical cumulative distribution function (cdf) and its sampling noise in Section 1. Using the limiting law of the scaled empirical-cdf error, we develop the computation and calibration of the Kolmogorov–Smirnov (KS) test (Section 2). Section 3 explains how the null reference changes when we fit parameters or change the comparison. We next replace the cdf curve by cell counts (Pearson, Section 4) and then by selected smooth features (Neyman, Section 5). *The choice of discrepancy determines which alternatives the test detects well.* At each step, we use the corresponding null law to set a rejection threshold. The final synthesis connects these choices to Week 9's Gaussian limits and score tests.

Prerequisites. Iid sampling, the multivariate CLT, the law of large numbers, continuous mapping and Slutsky's theorem, Taylor expansion, orthogonal projections, Week 6's empirical processes, and Week 9's local alternatives and score tests. We recall the empirical-cdf limits and Brownian bridge from Week 6 and use them to construct goodness-of-fit tests.

Reading. van der Vaart [1], §§17.1–17.3 and 17.5 for chi-square tests and fitted parameters, and §§19.1–19.3 for empirical distributions and goodness-of-fit statistics. Neyman [2] introduced the smooth-test construction. Rayner, Thas, and Best [3] develop its score-test interpretation and extensions. Sources and further reading appear at the end.

Contents

0 The testing problem and notation . . . 2
1 The empirical distribution and its noise . . . 2
2 The Kolmogorov–Smirnov test . . . 6
3 Extensions of KS and other cdf tests . . . 8
4 Pearson's chi-squared test . . . 12
5 Neyman's smooth test . . . 15
6 One discrepancy, three summaries . . . 17
Exercises . . . 17
Supplementary Material . . . 21

0 The testing problem and notation

Let $X_1, \dots, X_n$ be independent and identically distributed (iid) real observations with cumulative distribution function (cdf) $F(x) = \mathbb{P}(X_1 \le x)$. A *simple null* specifies the entire cdf:

$$H_0 : F = F_0, \qquad H_1 : F \neq F_0.$$

A *composite null* specifies a family $\mathcal{F}_0 = \{F_\theta : \theta \in \Theta \subset \mathbb{R}^d\}$ and asks whether $F \in \mathcal{F}_0$. For example, $N(0,1)$ is a simple null, whereas normality with unknown mean and variance is composite. In practice, to test normality with unknown mean and variance, we first estimate these parameters from the sample and compare the data with the fitted normal distribution. *Because we use the same data to fit and compare, parameter estimation changes how large a discrepancy we should expect even under normality.* We must account for this change when choosing the cutoff for rejecting normality. For example, if we obtain the cutoff by simulation, we must re-estimate the mean and variance in each simulated sample.

Recap. A test rejects when a statistic T_n is large. Its *level* is at most $\alpha \in (0,1)$ if its rejection probability is at most α for every null distribution. A sequence has *pointwise asymptotic level* α if the limiting upper rejection probability is at most α at each fixed null distribution. The composite asymptotic statements below are pointwise unless stated otherwise. A *p-value* is the null upper-tail probability of a statistic at least as large as its observed value. A large p-value means that this test did not detect a departure, but it does not establish that the model is true.

Write $P_n f = n^{-1}\sum_{i=1}^n f(X_i)$, $\mathbb{E}_F f = \int f\, dF$, and $\|g\|_\infty = \sup_x |g(x)|$. The symbols $\xrightarrow{\mathbb{P}}$, $\rightsquigarrow$, $o_{\mathbb{P}}(1)$, and $O_{\mathbb{P}}(1)$ denote convergence in probability, convergence in distribution, a term that tends to zero in probability, and a sequence bounded in probability. For vectors, $\|v\|^2 = v^{\mathrm{T}}v$. A symmetric matrix Π is an *orthogonal projection* if $\Pi^2 = \Pi$. Its rank is the dimension of the space onto which it projects.

For a positive integer r, χ^2_r is the distribution of $\sum_{j=1}^r Z_j^2$ for independent $Z_j \sim N(0,1)$; $q_{r,1-\alpha}$ is its $(1-\alpha)$ quantile. The noncentral law $\chi^2_r(\lambda)$ is the distribution of $\sum_{j=1}^r (Z_j + \mu_j)^2$ when $\sum_j \mu_j^2 = \lambda$. Its parameter λ measures squared signal after standardization. We keep the cell count k and smooth-test order m distinct throughout.

Remark 0.1. All observations are assumed iid unless a two-sample design is stated. Serial dependence, censoring, clustered observations, and residuals from a fitted regression require additional theory. Treating them as an iid sample can change the null distribution.

1 The empirical distribution and its noise

We recall the empirical cdf and its sampling error from Week 6. Here they supply the discrepancy and its noise reference for goodness-of-fit tests; our next step is to choose how to summarize that discrepancy.

1.1 A distribution estimate with no fitted parameters

Definition 1.1 (Empirical cdf)**.** The empirical cdf is $F_n(x) = n^{-1}\sum_{i=1}^n \mathbf{1}\{X_i \le x\}$. It is a right-continuous step function, with a jump of $1/n$ at each observation when there are no ties.

For each fixed x, the summands are Bernoulli with success probability $F(x)$. Thus

$$\mathbb{E}_F F_n(x) = F(x), \qquad \mathrm{Var}_F(F_n(x)) = \frac{F(x)\,(1-F(x))}{n}.$$

For two thresholds x, y, the indicators are nested, so

$$n\,\mathrm{Cov}_F(F_n(x), F_n(y)) = F(\min\{x,y\}) - F(x)F(y). \tag{1.1}$$

Errors at nearby thresholds are therefore dependent. Testing the whole cdf requires a joint limit, rather than a separate normal approximation at every x.

1.2 Reduction to uniformity

The null cdf can have any continuous shape. A common scale will let us use the same noise reference for all such fully specified nulls.

Proposition 1.2 (Probability integral transform)**.** *If $X \sim F_0$ and F_0 is continuous, then $U = F_0(X) \sim \mathrm{Unif}(0,1)$. Conversely, if $U \sim \mathrm{Unif}(0,1)$ and $F_0^{-1}(u) = \inf\{x : F_0(x) \geq u\}$, then $F_0^{-1}(U)$ has cdf F_0. Strict increase of F_0 is not required.*

Proof of Proposition 1.2. For $0 < u < 1$, define $x_u := \inf\{x \in \mathbb{R} : F_0(x) \geq u\}$. This specifies the left endpoint even when $F_0(x) = u$ has several solutions. The limits of a cdf at $\pm\infty$ ensure that x_u is finite, and continuity gives $F_0(x_u) = u$. By monotonicity and the definition of x_u,

$$\{F_0(X) < u\} = \{X < x_u\}, \qquad \mathbb{P}\{F_0(X) < u\} = F_0(x_u-) = u.$$

Since this holds for every $0 < u < 1$, $F_0(X)$ is uniform on $[0,1]$. For the converse, the quantile identity $\{F_0^{-1}(U) \leq x\} = \{U \leq F_0(x)\}$ holds for $U \in (0,1)$, so its probability is $F_0(x)$. □

For the transformed sample $U_i = F_0(X_i)$, write $G_n(u) = n^{-1}\sum_i \mathbf{1}\{U_i \leq u\}$. *Under a continuous simple null the U_i are iid uniform, and the distribution of a statistic formed from them no longer depends on F_0.* This makes *distribution-free* tests possible.

1.3 Uniform laws and a Brownian bridge

Recall the Brownian bridge from Week 6, §24. We now use it to describe the sampling noise that a goodness-of-fit test must distinguish from a model departure. A Gaussian process is one whose every finite vector of evaluations is jointly normal.

Definition 1.3 (Standard Brownian motion)**.** A one-dimensional standard Brownian motion $W = (W(t))_{0 \leq t \leq 1}$ is a real-valued stochastic process with $W(0) = 0$ almost surely, almost surely continuous sample paths, and the following properties:

(i) **Normal increments.** For every $0 \leq s < t \leq 1$,

$$W(t) - W(s) \sim N(0, t-s).$$

(ii) **Independent increments.** For any $0 \leq t_0 < t_1 < \cdots < t_r \leq 1$, the increments

$$W(t_1) - W(t_0), \ldots, W(t_r) - W(t_{r-1}) \quad \text{are independent.}$$

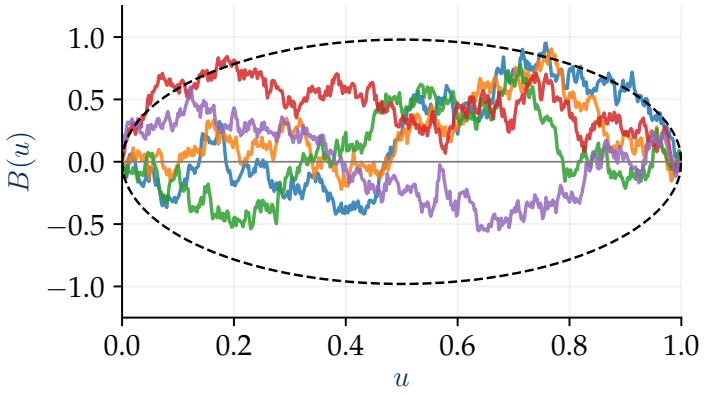


Figure 1: Independent simulated Brownian-bridge paths. Variability is largest near $u = 1/2$ and vanishes at both endpoints. The dashed curves are pointwise $\pm 1.96\sqrt{u(1-u)}$ guides, not simultaneous 95% bands.

The increment law depends only on the interval length; this is called *stationary increments*. The defining properties also make W a mean-zero Gaussian process with

$$\mathbb{E}W(t) = 0, \qquad \mathrm{Cov}(W(s), W(t)) = \min(s, t).$$

For the d-dimensional version with independent coordinates, the normal-increment law becomes $N_d(0, (t-s)I_d)$, where I_d is the identity matrix. We use the one-dimensional version below.

Definition 1.4 (Standard Brownian bridge)**.** Let W be standard Brownian motion on $[0, 1]$. The process $B(u) = W(u) - uW(1)$, $0 \le u \le 1$, is a standard Brownian bridge. It is a mean-zero Gaussian process with continuous paths and

$$B(0) = B(1) = 0, \qquad \mathrm{Cov}(B(s), B(t)) = \min(s, t) - st.$$

Subtracting $uW(1)$ pins the path to zero at $u = 1$. For uniform data, $G_n(0) = 0$ and $G_n(1) = 1$ almost surely, so the centered error $G_n(u) - u$ also vanishes at both endpoints. The covariance of $\sqrt{n}\{G_n(u) - u\}$ matches that of the bridge by (1.1).

Theorem 1.5 (Empirical-cdf limits and elementary concentration)**.** *For iid observations with cdf F, $\|F_n - F\|_\infty \to 0$ almost surely (Glivenko–Cantelli). For iid uniform observations,*

$$\alpha_n(u) = \sqrt{n}\{G_n(u) - u\} \rightsquigarrow B(u), \qquad 0 \le u \le 1,$$

in the space of bounded functions with the supremum norm (Donsker). In particular, continuous functionals for that norm converge in distribution. Also, for every cdf F and every $t > 0$,

$$\mathbb{E}_F\|F_n - F\|_\infty \le \frac{4}{\sqrt{n}}, \tag{1.2}$$

$$\mathbb{P}_F\left\{\|F_n - F\|_\infty > \frac{4}{\sqrt{n}} + t\right\} \le \exp(-2nt^2). \tag{1.3}$$

The finite-sample bounds require no continuity of F.

Proof of Theorem 1.5. For Glivenko–Cantelli, see the proof of Week 6, Theorem 3.1 (p. 3). The process limit is the uniform-cdf case of Week 6, Theorem 24.1 (p. 19); continuous mapping gives the assertion about continuous functionals. We now prove the finite-sample bounds.

Step 1: bound the expected error. Set $\Delta_n = \|F_n - F\|_\infty$. Right-continuity lets us take the supremum over rational thresholds, so Δ_n is measurable. Let $\epsilon_1, \dots, \epsilon_n$ be iid signs with probabilities $\mathbb{P}(\epsilon_i = 1) = \mathbb{P}(\epsilon_i = -1) = 1/2$, independent of the observations. Symmetrization, proved in Week 6, Section 5, gives

$$\mathbb{E}\Delta_n \le \frac{2}{n}\mathbb{E}\sup_x \left|\sum_{i=1}^n \epsilon_i \mathbf{1}\{X_i \le x\}\right|.$$

Condition on the observations and order them. As x increases, the sign sum runs through a subset of the partial sums $S_k = \sum_{j=1}^k \epsilon_j$, with $S_0 = 0$. Reordering the independent signs preserves their conditional law; ties only remove possible partial sums.

Lemma 1.6 (Maximum of a random-sign walk)**.** *Let $\epsilon_1, \dots, \epsilon_n$ be independent random signs with equal probabilities, and put $S_0 = 0$ and $S_k = \sum_{j=1}^k \epsilon_j$. Then*

$$\mathbb{E}\max_{0\le k\le n} |S_k| \le 2\sqrt{n}.$$

Applying this lemma to the ordered sign process yields

$$\mathbb{E}\Delta_n \le \frac{2}{n}\mathbb{E}\max_{0\le k\le n}|S_k| \le \frac{4}{\sqrt{n}},$$

which proves (1.2).

Step 2: control deviations from the mean. We use the following concentration lemma.

Lemma 1.7 (Bounded-differences concentration)**.** *Let $Z = f(X_1, \dots, X_n)$ be bounded and measurable, with independent X_i. Suppose replacing X_i while holding the other coordinates fixed changes f by at most $c_i \ge 0$. If $\sum_i c_i^2 > 0$, then, for every $t > 0$,*

$$\mathbb{P}(Z - \mathbb{E}Z > t) \le \exp\left(-\frac{2t^2}{\sum_i c_i^2}\right).$$

Replacing one observation changes $F_n(x)$ by at most $1/n$ for every x. The reverse triangle inequality for the supremum norm therefore changes Δ_n by at most $1/n$. Apply the lemma with $Z = \Delta_n$ and $c_i = 1/n$. Since $\sum_i c_i^2 = 1/n$ and $\mathbb{E}\Delta_n \le 4/\sqrt{n}$,

$$\mathbb{P}\left\{\Delta_n > \frac{4}{\sqrt{n}} + t\right\} \le \mathbb{P}\{\Delta_n > \mathbb{E}\Delta_n + t\} \le e^{-2nt^2},$$

which proves (1.3). Both supporting lemmas are proved in Appendix B.1. □

The process limit controls oscillations between thresholds as well as finite-dimensional fluctuations. Appendix B.2 explains why a finite-dimensional CLT alone is insufficient.

Remark 1.8 (Sharp DKW bound). The sharper Dvoretzky–Kiefer–Wolfowitz (DKW) inequality states that, for every cdf F and every $\varepsilon > 0$,

$$\mathbb{P}_F\{\|F_n - F\|_\infty > \varepsilon\} \le 2\exp(-2n\varepsilon^2). \tag{1.4}$$

The elementary proof already gives the $n^{-1/2}$ scale; sharp DKW improves the constants. Full proofs can be found at Massart (1990) and Reeve (2024), Sections 1–2 and Corollary 2 [4, 5]. Reeve proves a stronger one-sided bound; applying the bound to both tails gives (1.4).

Remark 1.9 (Intuition). The Brownian bridge describes the sampling noise of the whole empirical cdf after transformation to uniformity. To turn this random curve into a test, Section 2 summarizes it by its largest absolute deviation.

2 The Kolmogorov–Smirnov test

Week 6, Theorem 24.1 (p. 19) states the process convergence $\sqrt{n}(F_n - F) \rightsquigarrow B \circ F$ for a continuous cdf F. Here $(B \circ F)(x) = B(F(x))$. Section 30 (pp. 21–22) records the limiting distribution of $\sqrt{n}\|F_n - F\|_\infty$. Here we develop the KS test: compute the observed gap, then choose a rejection threshold under the specified null model.

2.1 Statistic and exact computation

For a fully specified continuous cdf F_0, define the two-sided KS statistic by

$$D_n = \sup_{x \in \mathbb{R}} |F_n(x) - F_0(x)| = \|F_n - F_0\|_\infty.$$

The two-sided KS test rejects for large D_n. To distinguish the two directions, also define

$$D_n^+ = \sup_x \{F_n(x) - F_0(x)\}, \qquad D_n^- = \sup_x \{F_0(x) - F_n(x)\}.$$

Then $D_n = \max(D_n^+, D_n^-)$. The superscripts describe the sign of the cdf discrepancy: D_n^+ detects excess probability below some threshold, often a shift toward smaller observations.

Proposition 2.1 (Order-statistic formula)**.** *Let $X_{(1)} \le \cdots \le X_{(n)}$ be the sorted observations and assume F_0 is continuous. The two one-sided components satisfy*

$$D_n^+ = \max_{1 \le i \le n} \left\{ \frac{i}{n} - F_0(X_{(i)}) \right\}, \qquad D_n^- = \max_{1 \le i \le n} \left\{ F_0(X_{(i)}) - \frac{i-1}{n} \right\}.$$

Consequently,

$$D_n = \sup_x |F_n(x) - F_0(x)| = \max\{D_n^+, D_n^-\} \tag{2.1}$$

Under H_0, the finite-sample distribution of D_n is the same for every continuous F_0.

Proof of Proposition 2.1. Between successive observations F_n is constant and F_0 is nondecreasing. For distinct observations,[1] fix $X_{(i)}$ and compare its two adjacent intervals in Figure 2. On the right, $F_n - F_0$ is nonincreasing, with maximum $i/n - F_0(X_{(i)})$ at the jump. On the left, $F_0 - F_n$ is nondecreasing, with supremum $F_0(X_{(i)}) - (i-1)/n$ at the left limit. Continuity of F_0 identifies these endpoint values. Maximizing over i gives the formulas for D_n^+ and D_n^-; both tail limits are zero. Since $|a| = \max(a, -a)$, taking the larger of the two directional maxima gives (2.1).

[1] The formula also holds with ties. If $X_{(a)} = \cdots = X_{(b)} = x$, then $F_n(x-) = (a-1)/n$ and $F_n(x) = b/n$. Within this block, the maximum for D_n^- uses a, and the maximum for D_n^+ uses b. Thus (2.1) still holds.

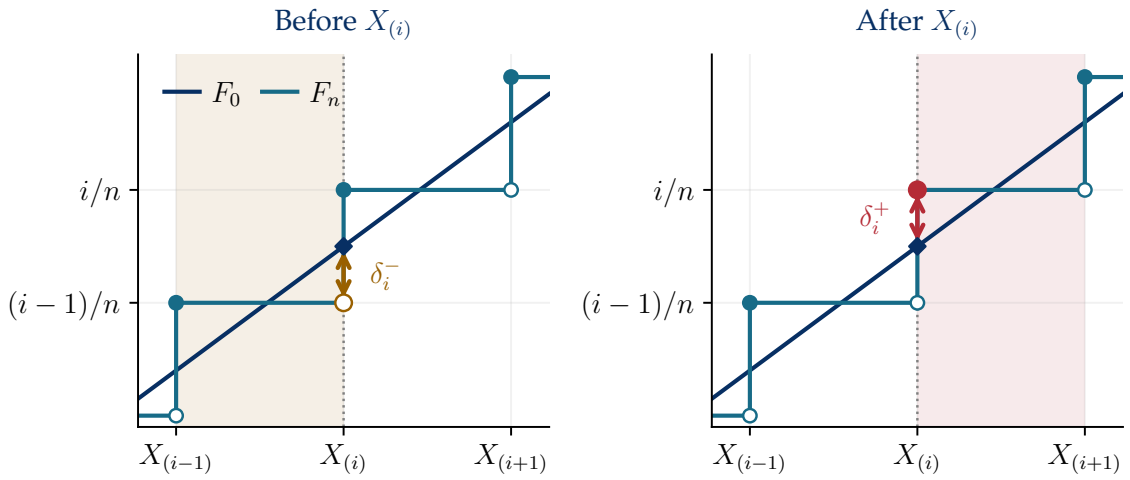


Figure 2: Two adjacent intervals at the same $X_{(i)}$, with $F_0(x) = x$ for illustration. Left: on $[X_{(i-1)}, X_{(i)})$, $F_n = (i-1)/n$ and $F_0 - F_n$ increases toward $\delta_i^- = F_0(X_{(i)}) - (i-1)/n$ at the open dot. Right: on $[X_{(i)}, X_{(i+1)})$, $F_n = i/n$ and $F_n - F_0$ decreases from $\delta_i^+ = i/n - F_0(X_{(i)})$ at the filled dot. These are local endpoint candidates; $D_n^{\pm} = \max_i \delta_i^{\pm}$.

Under H_0, the transformed ordered observations $F_0(X_{(i)})$ are ordered iid uniforms by Proposition 1.2. The formula therefore has a null distribution depending only on n.

□

Using only $|i/n - F_0(X_{(i)})|$ misses the left side of each jump and can understate the KS distance. Both sides matter. Ties in a truly continuous iid sample have probability zero.

2.2 The limiting law and a rejection threshold

The order-statistic formula computes the observed gap. To choose a cutoff, we need the following theorem.

Theorem 2.2 (KS null limit and consistency)**.** *For iid observations from a continuous, fully specified cdf F_0,*

$$\sqrt{n} D_n \rightsquigarrow K = \sup_{0 \le u \le 1} |B(u)|.$$

Here B is a standard Brownian bridge on $[0,1]$. Let c_α satisfy $\mathbb{P}(K > c_\alpha) = \alpha$. Rejecting when $\sqrt{n} D_n > c_\alpha$ has asymptotic level α and is consistent against every fixed cdf $F \neq F_0$.

Proof of Theorem 2.2. Under H_0, Proposition 1.2 gives iid uniform observations $U_i = F_0(X_i)$, with empirical cdf G_n, and thus $U_{(i)} = F_0(X_{(i)})$. Apply Proposition 2.1 first to the original sample with null cdf F_0, then to the transformed sample with the uniform null on $[0,1]$. The same ordered-data expression gives

$$D_n = \max_{1 \le i \le n} \max \left\{ \frac{i}{n} - U_{(i)},\ U_{(i)} - \frac{i-1}{n} \right\} = \sup_{0 \le u \le 1} |G_n(u) - u|.$$

Now define $\alpha_n(u) = \sqrt{n}\{G_n(u) - u\}$. The displayed identity gives $\sqrt{n} D_n = \|\alpha_n\|_\infty$. Theorem 1.5 states that the random functions α_n converge in distribution to B. The map $g \mapsto \|g\|_\infty$ is Lipschitz

with constant one, so continuous mapping gives $\sqrt{n}D_n \rightsquigarrow \|B\|_\infty$. Kolmogorov's (1933) distribution formula [6], stated as Lemma B.1 in Appendix B.3, gives, for $c > 0$,

$$\mathbb{P}(K \le c) = 1 - 2\sum_{j=1}^{\infty}(-1)^{j-1}e^{-2j^2c^2}. \tag{2.2}$$

The Portmanteau lemma (Week 9, Lemma 1.6(ii), p. 5) applies to (c_α, ∞) because continuity of the cdf of K gives $\mathbb{P}(K = c_\alpha) = 0$. Hence

$$\mathbb{P}(\sqrt{n}D_n > c_\alpha) \longrightarrow \mathbb{P}(K > c_\alpha) = \alpha.$$

For a fixed alternative put $\delta = \|F - F_0\|_\infty > 0$. The reverse triangle inequality gives $|D_n - \delta| \le \|F_n - F\|_\infty \to 0$ almost surely. Since $c_\alpha/\sqrt{n} \to 0$, rejection probability tends to one. □

Remark 2.3. For $\alpha = .05$, $c_\alpha \approx 1.35809864$. The upper-tail series in (2.2) gives an asymptotic p-value at $c = \sqrt{n}D_n$.

2.3 Finite-sample bands from DKW

The bridge gives a limiting reference. For finite-sample calibration, use the sharp DKW inequality in Remark 1.8. Under $H_0 : F = F_0$, (1.4) gives

$$\mathbb{P}_{F_0}(D_n > \varepsilon) \le 2\exp(-2n\varepsilon^2).$$

Setting the right-hand side equal to α gives the rejection threshold

$$\varepsilon_{n,\alpha} = \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

By (1.4), rejecting when $D_n > \varepsilon_{n,\alpha}$ has level at most α for any fully specified cdf, including a discrete cdf. This is a bound, not an assertion of exact size α. Likewise,

$$\big[\max\{0, F_n(x) - \varepsilon_{n,\alpha}\},\ \min\{1, F_n(x) + \varepsilon_{n,\alpha}\}\big], \qquad x \in \mathbb{R},$$

is a simultaneous confidence band with coverage at least $1 - \alpha$. At $\alpha = .05$, the DKW coefficient is $\sqrt{n}\varepsilon_{n,.05} \approx 1.35810152$, whereas the limiting KS cutoff is $c_{.05} \approx 1.35809864$. The DKW coefficient is slightly larger, by approximately 2.8764×10^{-6}; both round to 1.3581 at four decimal places.

3 Extensions of KS and other cdf tests

We now have a test for a continuous, fully specified null. Its construction separates the chosen discrepancy from its null reference. We use this distinction to change the direction or the samples being compared, summarize the cdf error differently, and account for fitted or discrete null models.

3.1 One-sided and two-sample tests

We first retain a direction of departure, then allow both cdfs to be estimated from data.

Proposition 3.1 (One-sided KS limits)**.** *For iid observations from a continuous, fully specified cdf F_0, let B be a standard Brownian bridge on $[0,1]$. Then*

$$\sqrt{n}D_n^+ \rightsquigarrow K^+ = \sup_{0\le u\le 1} B(u), \qquad \sqrt{n}D_n^- \rightsquigarrow K^- = \sup_{0\le u\le 1}\{-B(u)\}.$$

For the same bridge, $K = \sup_{0\le u\le 1}|B(u)| = \max\{K^+, K^-\}$. For $c>0$, $\mathbb{P}(K^+ > c) = \mathbb{P}(K^- > c) = e^{-2c^2}$. For a direction fixed in advance, rejecting when the corresponding scaled statistic exceeds $\sqrt{\log(1/\alpha)/2}$ has asymptotic level α.

Proof of Proposition 3.1. The probability transform and Theorem 1.5 give a uniform empirical process converging to B. Apply the continuous maps $g \mapsto \sup g$ and $g \mapsto \sup(-g)$. For the boundary law, let W be standard Brownian motion and φ the standard normal density. Reflecting a path after its first hit of c preserves its probability law and replaces its endpoint $y < c$ by $2c - y$. Thus the joint density of $W(1) = y$ and $\sup_{0\le u\le 1} W(u) \ge c$ is $\varphi(2c-y)$ for $y < c$. A standard bridge has the conditional law of W given $W(1) = 0$, so

$$\mathbb{P}(K^+ \ge c) = \frac{\varphi(2c)}{\varphi(0)} = e^{-2c^2}.$$

This tail is continuous, and $-B$ has the same law as B. The cutoff and rejection-probability limit follow from the Portmanteau lemma, as in the proof of Theorem 2.2. □

Remark 3.2. At $\alpha = .05$, the one-sided cutoff is approximately 1.2239.

Proposition 3.3 (Two-sample KS limit)**.** *Let $X_1,\dots,X_n \overset{\text{iid}}{\sim} F$ and $Y_1,\dots,Y_\ell \overset{\text{iid}}{\sim} H$ be independent samples, with empirical cdfs F_n and H_ℓ. Under $H_0 : F = H$ with a common continuous cdf, as $n,\ell\to\infty$ with $n/(n+\ell)\to\tau\in(0,1)$,*

$$\sqrt{\frac{n\ell}{n+\ell}}\|F_n - H_\ell\|_\infty \rightsquigarrow K = \sup_{0\le u\le 1}|B(u)|, \tag{3.1}$$

where B is a standard Brownian bridge. Thus the limiting cutoff is the same c_α as in Theorem 2.2.

Proof of Proposition 3.3. Transform both samples by their common cdf. Their independent uniform empirical processes converge jointly to independent standard bridges B_1, B_2. The scaled difference converges to $\sqrt{1-\tau}B_1 - \sqrt{\tau}B_2$, a centered continuous Gaussian process with covariance

$$\{(1-\tau)+\tau\}\{\min(u,v) - uv\} = \min(u,v) - uv.$$

It is therefore a standard bridge. Apply the supremum map; continuity of the cdf of K gives the cutoff claim. □

Exercise 4 develops finite-sample permutation calibration, including tied observations.

3.2 Integrated discrepancies

KS retains the largest cdf error. We can instead accumulate squared errors across thresholds. Let F_0 be a continuous, fully specified null cdf and put $U_{(i)} = F_0(X_{(i)})$. The Cramér–von Mises (CvM)

and Anderson–Darling (AD) statistics are

$$W_n^2 = n\int (F_n - F_0)^2\, dF_0 = \frac{1}{12n} + \sum_{i=1}^{n}\left(U_{(i)} - \frac{2i-1}{2n}\right)^2, \tag{3.2}$$

$$A_n^2 = n\int \frac{(F_n - F_0)^2}{F_0(1-F_0)}\, dF_0 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\{\log U_{(i)} + \log(1 - U_{(n+1-i)})\}. \tag{3.3}$$

The ordered-data formulas hold almost surely under the null. CvM accumulates squared cdf errors across the distribution, while AD gives larger weights near the two tails.

Proposition 3.4 (CvM and AD null limits)**.** *For iid observations from the continuous cdf F_0, let B be a standard Brownian bridge on $[0,1]$. The statistics in (3.2)–(3.3) satisfy*

$$W_n^2 \rightsquigarrow \int_0^1 B(u)^2\, du, \qquad A_n^2 \rightsquigarrow \int_0^1 \frac{B(u)^2}{u(1-u)}\, du.$$

Both limiting integrals are finite almost surely.

Proof of Proposition 3.4. Let G_n be the empirical cdf of the iid uniforms $U_i = F_0(X_i)$ and put $\alpha_n(u) = \sqrt{n}\{G_n(u) - u\}$. The probability transform gives $W_n^2 = \int_0^1 \alpha_n^2$ and $A_n^2 = \int_0^1 \alpha_n^2/\{u(1-u)\}$. On $[U_{(i)}, U_{(i+1)})$, $G_n = i/n$, where $U_{(0)} = 0$ and $U_{(n+1)} = 1$. Integrating on these intervals and collecting endpoint terms gives (3.2) and (3.3); for AD use

$$\frac{(p-u)^2}{u(1-u)} = \frac{p^2}{u} + \frac{(1-p)^2}{1-u} - 1, \qquad p = i/n.$$

Theorem 1.5 gives $\alpha_n \rightsquigarrow B$. Define $\Phi(g) = \int_0^1 g(u)^2\, du$. For bounded measurable paths g, h,

$$|\Phi(g) - \Phi(h)| \le \|g - h\|_\infty\big(\|g\|_\infty + \|h\|_\infty\big),$$

Thus Φ is continuous in the supremum norm. Applying the metric-space extension of Week 9, Lemma G.1(i) (see Lemma A.1) gives $W_n^2 = \Phi(\alpha_n) \rightsquigarrow \Phi(B)$.

CMT imposes no integrability requirement. For AD, define

$$\Psi(g) = \int_0^1 \frac{g(u)^2}{u(1-u)}\, du, \qquad v_m(u) = m^{-1/2}\mathbf{1}\{e^{-2m} \le u < e^{-m}\}, \quad m \ge 1.$$

This map is not continuous in the supremum norm: $\|v_m\|_\infty = m^{-1/2} \to 0$, whereas

$$1 \le \Psi(v_m) = \frac{1}{m}\int_{e^{-2m}}^{e^{-m}} \frac{du}{u(1-u)} \le \frac{1}{1-e^{-m}} \longrightarrow 1.$$

For any path g with $\Psi(g) < \infty$, Cauchy–Schwarz on the shrinking support of v_m shows that the cross term in $\Psi(g + v_m)$ tends to zero. Hence $\Psi(g + v_m) - \Psi(g) \to 1$. Thus continuity fails even at the bridge paths, whose weighted integral is finite almost surely, as verified below.

We therefore apply CMT after truncation: for $0 < \delta < 1/2$, define

$$\Psi_\delta(g) = \int_\delta^{1-\delta} \frac{g(u)^2}{u(1-u)}\, du.$$

The weight is bounded on this interval, so the same continuity argument gives $\Psi_\delta(\alpha_n) \rightsquigarrow \Psi_\delta(B)$. To restore the endpoints, put $E_\delta = (0,\delta) \cup (1-\delta, 1)$. Since $\mathbb{E}\alpha_n(u)^2 = \mathbb{E}B(u)^2 = u(1-u)$, Tonelli's theorem gives

$$\mathbb{E}\int_{E_\delta} \frac{\alpha_n(u)^2}{u(1-u)}\,du = \mathbb{E}\int_{E_\delta} \frac{B(u)^2}{u(1-u)}\,du = 2\delta.$$

Markov's inequality makes both omitted endpoint terms negligible as $\delta \downarrow 0$, uniformly in n for the empirical process. Taking $n \to \infty$ first and then $\delta \downarrow 0$ therefore gives the AD limit [7].

Finally, Tonelli's theorem gives

$$\mathbb{E}\int_0^1 B(u)^2\,du = 1/6, \qquad \mathbb{E}\int_0^1 \frac{B(u)^2}{u(1-u)}\,du = 1.$$

Thus both limiting integrals are finite almost surely. □

3.3 Fitting a null model changes the process

The normality example in Section 0 raised a different issue: the reference cdf was fitted from the same data. We now describe how fitting changes the whole discrepancy process.

Setup and conditions. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F_{\theta_0}$ for an unknown interior $\theta_0 \in \mathbb{R}^d$, with fixed d and continuous F_{θ_0}. Suppose an estimator has a fixed measurable influence function $\psi : \mathbb{R} \to \mathbb{R}^d$ satisfying

$$\sqrt{n}(\widehat{\theta} - \theta_0) = Z_n + o_{\mathbb{P}}(1), \qquad Z_n = n^{-1/2}\sum_{i=1}^n \psi(X_i), \quad \mathbb{E}\psi = 0, \quad \mathbb{E}\|\psi\|^2 < \infty.$$

Let $\dot{F}(x) = \dot{F}_{\theta_0}(x)$ be the cdf derivative column vector, with $\sup_x \|\dot{F}(x)\| < \infty$. Assume

$$\sup_x \sqrt{n}\left|F_{\widehat{\theta}}(x) - F_{\theta_0}(x) - \dot{F}(x)^{\mathrm{T}}(\widehat{\theta} - \theta_0)\right| = o_{\mathbb{P}}(1).$$

These iid and moment conditions give joint convergence $(\sqrt{n}(F_n - F_{\theta_0}), Z_n) \rightsquigarrow (\mathbb{G}, Z)$ in the supremum norm and $\mathbb{R}^d$; see Remark 3.6. The limit is centered Gaussian, and $\mathbb{G}$ has the law of $B \circ F_{\theta_0}$ for a standard Brownian bridge B.

Proposition 3.5 (Fitted empirical-process limit)**.** *Under the preceding iid setup, estimator expansion, and uniform cdf expansion,*

$$\sqrt{n}\{F_n(x) - F_{\widehat{\theta}}(x)\} \rightsquigarrow \mathbb{G}(x) - \dot{F}(x)^{\mathrm{T}}Z \tag{3.4}$$

as random functions of x, with convergence in the supremum norm. In particular,

$$\sqrt{n}\|F_n - F_{\widehat{\theta}}\|_\infty \rightsquigarrow \sup_x |\mathbb{G}(x) - \dot{F}(x)^{\mathrm{T}}Z|.$$

Proof of Proposition 3.5. Subtract the uniform cdf expansion and substitute the estimator expansion to obtain

$$\sqrt{n}\{F_n(x) - F_{\widehat{\theta}}(x)\} = \sqrt{n}\{F_n(x) - F_{\theta_0}(x)\} - \dot{F}(x)^{\mathrm{T}}Z_n + o_{\mathbb{P}}(1).$$

Boundedness of $\dot{F}$ keeps the remainder uniform in x. Joint convergence and continuous mapping give (3.4). Applying the supremum map gives the fitted KS limit. □

Fitting parameters generally changes the KS reference distribution. With a fully specified cdf, the process limit $\mathbb{G}(x)$ has the law of $B(F_{\theta_0}(x))$. Fitting replaces it by $\mathbb{G}(x) - \dot{F}(x)^{\mathrm{T}} Z$. This is still a centered Gaussian process, but its covariance generally differs from that of $B \circ F_{\theta_0}$. Thus the fitted KS limit generally has a different distribution from $K = \sup_{0 \le u \le 1} |B(u)|$, and its critical values must account for the fitting procedure.

Remark 3.6 (Joint convergence). In the setup of Proposition 3.5, Theorem 1.5 gives asymptotic tightness of $\sqrt{n}(F_n - F_{\theta_0})$, and the multivariate CLT gives tightness of Z_n. The pair is therefore asymptotically tight, so every subsequence has a weakly convergent further subsequence. For any finite set of thresholds, the multivariate CLT applied jointly to their indicators and $\psi(X_i)$ identifies the finite-dimensional limits as those of $(\mathbb{G}, Z)$. These uniquely determine every subsequential limit, yielding the stated joint convergence.

Exercise 6 derives the efficient-MLE covariance and applies it to an exponential model.

3.4 Discrete nulls and randomized transforms

The continuous-null argument also used uniformity of $F_0(X)$. For discrete data, we must revisit this step. If F_0 has jumps, $F_0(X)$ is generally not uniform. The statistic $D_n = \|F_n - F_0\|_\infty$ is still meaningful, but its null law depends on the atom probabilities. Simulate directly from a fully specified discrete F_0 and use a discrete exact method.

A randomized transform is also possible. Let $V_i \overset{\text{iid}}{\sim} \mathrm{Unif}(0,1)$ be independent of the data and put

$$U_i = F_0(X_i-) + V_i\{F_0(X_i) - F_0(X_i-)\}, \qquad F_0(x-) = \mathbb{P}_{F_0}(X < x).$$

Under the simple null these U_i are iid uniform. Each atom is spread uniformly over its cdf jump interval. The resulting uniformity test can be conservative when calibrated by a bound such as DKW. With exact level-α calibration, its rejection probability is α, averaging over both the data and the added randomization.

4 Pearson's chi-squared test

The preceding tests compare cumulative probabilities along an ordered scale. For categorical outcomes or a meaningful partition, we can compare cell probabilities directly. This replaces the random cdf curve by a finite vector of count discrepancies. Pearson's test groups outcomes into k mutually exclusive, exhaustive cells $C_1, \dots, C_k$, which may be categories or intervals. Write

$$N_j = \sum_{i=1}^{n} \mathbf{1}\{X_i \in C_j\}, \qquad \widehat{p}_j = N_j/n, \qquad p_j = \mathbb{P}_{F_0}(X \in C_j).$$

Under the simple null, $(N_1, \dots, N_k)$ is multinomial with parameters n and $p = (p_1, \dots, p_k)^{\mathrm{T}}$.

4.1 The simple null and the missing degree of freedom

Assumption 4.1 (Fixed cells with positive probabilities)**.** The cells are fixed before observing the data, their number $k \ge 2$ is fixed as $n \to \infty$, and every null probability p_j is strictly positive. Also $\sum_{j=1}^{k} p_j = 1$.

The positivity condition allows standardization by $\sqrt{p_j}$ and ensures $np_j \to \infty$. These are the fixed-multinomial conditions used in [1, Chapter 17]. A cell with zero null probability needs separate treatment: any observation in it contradicts the null. Growing cell counts and probabilities tending to zero require a different limit argument.

Theorem 4.2 (Pearson null law)**.** *Under Assumption 4.1 and the iid simple null,*

$$X_n^2 = \sum_{j=1}^{k} \frac{(N_j - np_j)^2}{np_j} = n \sum_{j=1}^{k} \frac{(\widehat{p}_j - p_j)^2}{p_j} \rightsquigarrow \chi^2_{k-1}. \tag{4.1}$$

Thus $X_n^2 > q_{k-1,1-\alpha}$ is an asymptotic level-α rule.

Proof of Theorem 4.2. Set $D = \operatorname{diag}(p_1, \ldots, p_k)$ and $t = (\sqrt{p_1}, \ldots, \sqrt{p_k})^{\mathrm{T}}$, so $t^{\mathrm{T}}t = 1$. Let Y_i be the cell-indicator vector with jth coordinate $Y_{ij} = \mathbf{1}\{X_i \in C_j\}$. Then $\widehat{p} = n^{-1}\sum_i Y_i$ and $\mathbb{E}Y_i = p$. Since the cells are mutually exclusive, $\mathbb{E}(Y_{ij}Y_{i\ell}) = p_j\mathbf{1}\{j = \ell\}$, so

$$\operatorname{Cov}(Y_i) = D - pp^{\mathrm{T}}.$$

The bounded iid vectors $D^{-1/2}(Y_i - p)$ therefore have mean zero and covariance

$$D^{-1/2}(D - pp^{\mathrm{T}})D^{-1/2} = I_k - tt^{\mathrm{T}},$$

where $D^{-1/2}p = t$. The multivariate CLT gives

$$Z_n = \sqrt{n}D^{-1/2}(\widehat{p} - p) \rightsquigarrow Z \sim N_k(0, I_k - tt^{\mathrm{T}}).$$

The matrix $I_k - tt^{\mathrm{T}}$ sends t to zero and fixes every vector orthogonal to t, so it is the orthogonal projection onto the $(k-1)$-dimensional space $t^{\perp}$. The Gaussian projection lemma, Lemma A.2, gives $\|Z\|^2 \sim \chi^2_{k-1}$. Since $X_n^2 = \|Z_n\|^2$, continuous mapping proves the result. Continuity of the chi-squared cdf gives the level statement. □

The constraint $\sum_j (N_j - np_j) = 0$ removes one random direction. There are k parameters, but only $k-1$ independent/free parameters. This geometric argument explains the degrees of freedom.

4.2 Estimating cell probabilities

Our goal is the fitted Pearson limit in Theorem 4.3, which determines the asymptotic critical values after estimating parameters from the cell counts. Under its regularity conditions, fitting d parameters changes the reference law from χ^2_{k-1} to χ^2_{k-1-d}. We analyze how fitting changes the cell-frequency errors to explain this reduction in degrees of freedom. Suppose the null probabilities form a smooth model $p(\theta) \in (0,1)^k$, with an interior $\theta_0 \in \mathbb{R}^d$ and $d < k-1$. Let $J = \partial p(\theta_0)/\partial\theta^{\mathrm{T}}$ be the $k \times d$ derivative matrix. Write

$$D = \operatorname{diag}(p(\theta_0)), \quad t = \sqrt{p(\theta_0)}, \quad A = D^{-1/2}J, \quad I_g = A^{\mathrm{T}}A.$$

The square root in t is componentwise. For a small parameter increment $h \in \mathbb{R}^d$, $p(\theta_0 + h) - p(\theta_0) = Jh + o(\|h\|)$. Hence

$$D^{-1/2}\{p(\theta_0 + h) - p(\theta_0)\} = Ah + o(\|h\|).$$

The columns of J give first-order changes in the original probabilities; the columns of A give the same changes after dividing cell j by $\sqrt{p_j(\theta_0)}$. Pearson's statistic uses these standardized coordinates. A small parameter change can therefore explain linear combinations of the columns of A, which are patterns of positive and negative errors across the cells. Under the grouped-estimation conditions below, fitting removes the component in this span to first order, and the test uses the remaining component. Since the probabilities sum to one, $t^{\mathrm{T}}A = \mathbf{1}^{\mathrm{T}}J = 0$.

Theorem 4.3 (Fitted Pearson law for grouped estimation)**.** *Assume fixed positive cell probabilities, differentiability of p at θ_0, $\operatorname{rank} J = d$, and consistency of an estimator $\widehat{\theta}$ with expansion*

$$\sqrt{n}(\widehat{\theta} - \theta_0) = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}Z_n + o_{\mathbb{P}}(1), \qquad Z_n = \sqrt{n}D^{-1/2}\{\widehat{p} - p(\theta_0)\}. \tag{4.2}$$

Then

$$X^2_{n,\mathrm{fit}} = n\sum_{j=1}^{k}\frac{\{\widehat{p}_j - p_j(\widehat{\theta})\}^2}{p_j(\widehat{\theta})} \rightsquigarrow \chi^2_{k-1-d}.$$

Regular grouped multinomial MLEs satisfy (4.2); so do regular minimum-Pearson-distance estimators. Here grouped *means that estimation uses the cell counts.*

Proof of Theorem 4.3. Taylor expansion and (4.2) give

$$\sqrt{n}D^{-1/2}\{\widehat{p} - p(\widehat{\theta})\} = (I_k - P_A)Z_n + o_{\mathbb{P}}(1), \quad P_A = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}.$$

By Theorem 4.2's CLT, the limiting covariance is

$$(I_k - P_A)(I_k - tt^{\mathrm{T}})(I_k - P_A) = I_k - tt^{\mathrm{T}} - P_A.$$

The equality uses $P_A t = 0$. This is a projection of rank $k - 1 - d$. Consistency and positivity permit replacement of denominators by their limits, and Lemma A.2 finishes the proof. □

Remark 4.4 (Intuition). One direction is removed because probabilities sum to one. *Fitting d identifiable grouped parameters removes d more directions.* The remaining variation measures departures that the fitted model cannot explain.

The estimator condition matters. An MLE from the original ungrouped observations generally has a different information matrix and need not satisfy (4.2). Its plug-in Pearson statistic need not have exactly $k - 1 - d$ chi-squared degrees of freedom. Appendix C derives the covariance for that case. Use an appropriate derived law or a bootstrap that repeats fitting and grouping; do not subtract d merely because d parameters were estimated. Random fitted-quantile cell boundaries also fall outside this fixed-cell theorem.

4.3 Local power and the information retained by cells

The null law determines the cutoff. To understand which departures the cells retain, we next examine how alternatives shift the residual vector. Let D, A, P_A be as in Theorem 4.3 at the reference null.

Proposition 4.5 (Multinomial local alternatives)**.** *Keep k fixed and $p_j > 0$. Suppose the observations*

fall into the cells with probabilities

$$p_n = p + h/\sqrt{n}, \qquad \mathbf{1}^{\mathrm{T}} h = 0, \quad h \in \mathbb{R}^k \text{ fixed}.$$

These probabilities are positive for all sufficiently large n. Then the simple-null Pearson statistic in (4.1) satisfies

$$X_n^2 \rightsquigarrow \chi^2_{k-1}(\lambda), \qquad \lambda = h^{\mathrm{T}} D^{-1} h = \sum_j h_j^2/p_j.$$

Under a regular grouped fit whose projection expansion also holds along p_n, the fitted limit has $k-1-d$ degrees of freedom and

$$\lambda_{\mathrm{fit}} = \|(I_k - P_A)D^{-1/2}h\|^2.$$

Proof of Proposition 4.5. The bounded cell-indicator vectors satisfy the triangular-array CLT. Thus $Z_n \rightsquigarrow N_k(D^{-1/2}h, I_k - tt^{\mathrm{T}})$. Its mean is orthogonal to t because $t^{\mathrm{T}} D^{-1/2} h = \mathbf{1}^{\mathrm{T}} h = 0$. Apply the noncentral part of Lemma A.2. For the fitted case apply $I_k - P_A$ to the same limit; the resulting mean is in the range of $I_k - tt^{\mathrm{T}} - P_A$. □

The limiting power is $\mathbb{P}\{\chi^2_r(\lambda) > q_{r,1-\alpha}\}$ with the relevant r and λ. This is Week 9's Gaussian-shift argument in a finite-dimensional residual space. For a fixed alternative cell vector $q \neq p$, the law of large numbers gives $X_n^2/n \xrightarrow{\mathbb{P}} \sum_j (q_j - p_j)^2/p_j > 0$, so the test is consistent. But a fixed partition cannot detect a change within cells that preserves all cell probabilities. The test addresses the grouped distribution. Making cells finer can retain more information, while also making calibration more demanding.

5 Neyman's smooth test

Pearson's test chooses which discrepancies to retain through its cell boundaries. Neyman's construction [2] instead averages chosen functions of $U_i = F_0(X_i)$. Under the null, U_i is uniform on $[0,1]$. A *departure from uniformity* means an alternative law for U_i that differs from this uniform law. With the polynomial features used below, the alternative densities in Exercise 8 are smooth functions of u, with derivatives of every order. The test compares the feature averages with their null values. Throughout this section the null F_0 is continuous and fully specified, unless a fitted extension is explicitly discussed.

5.1 Orthonormal features and interpretable departures

For functions on $[0,1]$, use the inner product $\langle f, g\rangle = \int_0^1 f(u)g(u)\,du$. Choose bounded functions $\phi_1, \ldots, \phi_m$, where m is fixed before seeing the data, such that

$$\int_0^1 \phi_j(u)\,du = 0, \qquad \int_0^1 \phi_j(u)\phi_l(u)\,du = \mathbf{1}\{j = l\}. \tag{5.1}$$

The first normalized shifted Legendre polynomials are

$$\begin{aligned}
\phi_1(u) &= \sqrt{3}(2u-1),\\
\phi_2(u) &= \sqrt{5}(6u^2 - 6u + 1),\\
\phi_3(u) &= \sqrt{7}(20u^3 - 30u^2 + 12u - 1).
\end{aligned}$$

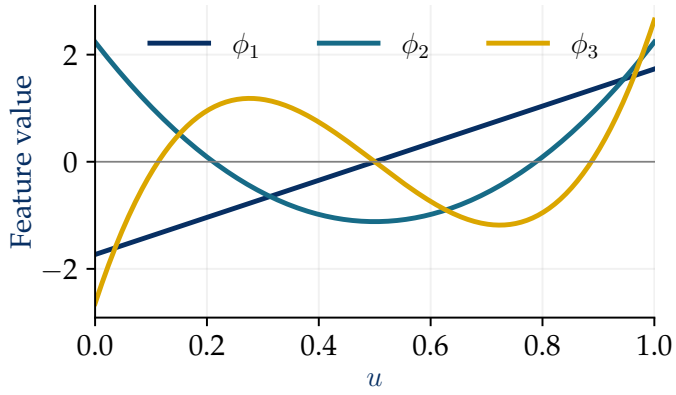


Figure 3: The first three normalized shifted Legendre features. Their empirical averages measure different smooth departures from uniformity.

They have mean zero and unit variance under uniformity, and their cross covariances vanish. These polynomials concern the transformed values U, not moments of the original X.

The first feature detects a change in the mean of U. The second detects a center-versus-ends departure after accounting for the first feature: indeed, $6u^2 - 6u + 1 = 6(u - 1/2)^2 - 1/2$. The third detects a higher-order asymmetric pattern. These descriptions explain sensitivity; they do not identify a unique original-data location, scale, or skewness parameter for an arbitrary F_0.

Definition 5.1 (Order-m smooth statistic)**.** Write $\phi(u) = (\phi_1(u), \dots, \phi_m(u))^{\mathrm{T}}$ and

$$Z_{n,j} = n^{-1/2} \sum_{i=1}^{n} \phi_j(U_i), \qquad Z_n = (Z_{n,1}, \dots, Z_{n,m})^{\mathrm{T}}.$$

Neyman's order-m smooth statistic is $S_{n,m} = \sum_{j=1}^{m} Z_{n,j}^2 = \|Z_n\|^2$.

Theorem 5.2 (Smooth-test null law)**.** *Under iid sampling from a continuous, fully specified F_0, for fixed bounded features satisfying (5.1),*

$$Z_n \rightsquigarrow N_m(0, I_m), \qquad S_{n,m} \rightsquigarrow \chi^2_m.$$

The rule $S_{n,m} > q_{m,1-\alpha}$ has asymptotic level α.

Proof of Theorem 5.2. The probability transform gives iid uniform U_i. By (5.1), $\mathbb{E}\phi(U_i) = 0$ and $\mathrm{Cov}(\phi(U_i)) = I_m$. The multivariate CLT gives the vector limit. Continuous mapping gives its squared norm, and continuity of the chi-squared distribution gives the rejection probability limit. □

Exercises 8 and 9 develop the score-test interpretation, local power, and choice of features. We now compare the distributional discrepancies retained by KS, Pearson, and smooth tests.

6 One discrepancy, three summaries

We have moved from cumulative errors to cell counts and then to smooth-feature averages. The three tests retain different parts of the same empirical discrepancy. Write $P_0 f = \mathbb{E}_{F_0} f$ for the null expectation. Then $\sqrt{n}(P_n - P_0)f$ measures the discrepancy for a chosen test function f. For $f_x = \mathbf{1}\{X \le x\}$ it is the cdf process. For cell indicators it is a multinomial vector. For $f_j = \phi_j(F_0(X))$ it is the smooth-feature vector. The statistical question is which functions should carry the signal.

Method	Discrepancy retained	Simple-null reference	Main design choice
KS	All cumulative thresholds; largest absolute gap	$\sup \lvert B \rvert$ after $\sqrt{n}$ scaling	One- or two-sided rule
Pearson	k cell probabilities	χ^2_{k-1}	Fixed partition
Smooth	m orthonormal feature means	χ^2_m	Basis and fixed order

There is also an exact link between Pearson and feature tests. Let $e_1, \dots, e_{k-1}$ be an orthonormal basis of the subspace perpendicular to $t = \sqrt{p}$, and define the piecewise constant features

$$\psi_l(x) = \sum_{j=1}^{k} \frac{e_{l,j}}{\sqrt{p_j}} \mathbf{1}\{x \in C_j\}, \qquad l = 1, \dots, k-1.$$

They have mean zero and identity covariance under the null. Because $Z_n = \sqrt{n} D^{-1/2}(\widehat{p} - p)$ lies in $t^\perp$ exactly, the Pearson statistic X_n^2 defined in (4.1) satisfies

$$X_n^2 = \sum_{l=1}^{k-1} \left\{ \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_l(X_i) \right\}^2. \tag{6.1}$$

Pearson is therefore an orthonormal-feature test with step functions; Neyman replaces these by smooth functions chosen to describe plausible departures. KS uses an entire threshold class and takes a maximum rather than a finite sum of squares.

A practical choice. Use KS when cumulative deviations are a useful general summary for univariate continuous data. Use Pearson when the recorded outcomes are categories or a partition is scientifically meaningful. Use a smooth test when a small set of shape departures has a clear interpretation. CvM or AD can suit an integrated discrepancy, with AD adding tail weight. These choices concern sensitivity and data structure; no method is uniformly most powerful over all departures.

Exercises

The exercises revisit the same construction: choose and compute a discrepancy, explain its null reference, and determine what changes after fitting parameters or selecting features.

Exercise 1 (Both sides of a jump)**.** Consider the ordered uniform-transformed observations

$$.04,\ .12,\ .27,\ .39,\ .49,\ .61,\ .72,\ .93.$$

Calculate D_n^+, D_n^-, and $\sqrt{n}D_n$. Explain why checking only $|i/n - U_{(i)}|$ is not a valid general formula, even if it happens to give the maximum in this example.

Exercise 2 (A simultaneous band)**.** Use DKW to construct a 95% confidence band for a cdf with $n = 200$. Is the coverage exactly 95%? Does continuity of the cdf matter for this bound?

Exercise 3 (Two-sample scaling)**.** There are $n = 80$ observations in one independent sample and $\ell = 120$ in another. Under a common continuous cdf, derive the asymptotic 5% cutoff for $\|F_n - H_\ell\|_\infty$. Explain the factor in (3.1) from the two variances.

Exercise 4 (Finite-sample permutation calibration)**.** Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_\ell \sim H$ be independent samples, iid within each group. Put $T_{\text{obs}} = \|F_n - H_\ell\|_\infty$. Pool the $N = n + \ell$ observations and let Π contain all $L = \binom{N}{n}$ allocations of n observation indices to the first group. Include the observed allocation and count by indices even when values coincide. For each allocation π, recompute both empirical cdfs and their distance T^π. Define

$$p_{\text{perm}} = \frac{1}{L} \sum_{\pi \in \Pi} \mathbf{1}\{T^\pi \geq T_{\text{obs}}\}.$$

(a) Under $H_0 : F = H$, prove that for every $0 < \alpha < 1$,

$$\mathbb{P}(p_{\text{perm}} \leq \alpha \mid \text{pooled unordered sample}) \leq \alpha.$$

Explain why this remains valid for discrete distributions and ties.

(b) For $n = \ell = 2$ with observed groups $\{1, 2\}$ and $\{3, 4\}$, list the six permutation distances and calculate the p-value. Explain the use of $\geq$ and the inclusion of the observed allocation.

Exercise 5 (Pearson's projection)**.** For a simple multinomial null, show $t^{\mathrm{T}} Z_n = 0$ exactly and verify $(I_k - tt^{\mathrm{T}})^2 = I_k - tt^{\mathrm{T}}$.

Exercise 6 (Efficient MLE covariance and an exponential model)**.** Use the setup of Proposition 3.5. Suppose $\psi = I^{-1}s$, where $s = s_{\theta_0}(X)$ is a mean-zero score and $I = \mathbb{E}(ss^{\mathrm{T}})$ is nonsingular Fisher information. Assume $\dot{F}(x) = \mathbb{E}[\mathbf{1}\{X \leq x\}s(X)]$, and write $H(x) = \mathbb{G}(x) - \dot{F}(x)^{\mathrm{T}} Z$ for the fitted-process limit.

(a) Derive $\mathrm{Var}(Z)$ and $\mathrm{Cov}(\mathbb{G}(x), Z)$ from the joint CLT. Show that

$$\mathrm{Cov}(H(x), H(y)) = F_{\theta_0}(\min\{x, y\}) - F_{\theta_0}(x)F_{\theta_0}(y) - \dot{F}(x)^{\mathrm{T}} I^{-1} \dot{F}(y). \tag{6.2}$$

(b) For $F_\lambda(x) = 1 - e^{-\lambda x}$, $x \geq 0$, $\lambda > 0$, find the MLE, score, Fisher information, and cdf derivative. Compute the covariance correction in part (a), and explain why ordinary simple-null KS tables do not calibrate the fitted statistic.

Exercise 7 (Fitted exponential calibration)**.** For the exponential model in Exercise 6, describe a Monte Carlo test with unknown rate using the fitted KS statistic. Prove that its null distribution does not depend on the rate. Explain why simulating a new sample but keeping the original

fitted rate inside the replicate statistic gives the wrong reference law.

Exercise 8 (Neyman's statistic as a score test)**.** Use the fixed bounded feature vector ϕ satisfying (5.1). Consider the density family [3]

$$g_\eta(u) = \exp\{\eta^{\mathrm{T}}\phi(u) - a(\eta)\}, \quad a(\eta) = \log \int_0^1 e^{\eta^{\mathrm{T}}\phi(v)}\, dv, \qquad \eta \in \mathbb{R}^m. \tag{6.3}$$

Write $\mathbb{E}_\eta$ for expectation under g_η.

(a) Show that g_η integrates to one and that $\eta = 0$ gives the uniform density. With only $\eta_1 > 0$ and $\phi_1(u) = \sqrt{3}(2u-1)$, describe how the density changes with u.

(b) Let $M = \sup_{u\in[0,1]} \|\phi(u)\|$. On a ball $\|\eta\| \le r$, bound the exponential integrand and its first two derivatives. Use these bounds to justify differentiation under the integral. Express $\nabla a(\eta)$ and $\nabla^2 a(\eta)$ as a mean and a covariance under g_η, and evaluate them at zero. State which feature condition gives each value.

(c) For iid observations $U_1, \dots, U_n \sim g_\eta$, derive the log likelihood, total score, and total Fisher information. Show that Rao's score statistic for testing $\eta = 0$ equals $S_{n,m}$.

(d) If $U_i = F_0(X_i)$ and the continuous null cdf F_0 has density f_0, express the corresponding alternative density on the original X scale.

Exercise 9 (Local power and the choice of order)**.** Use the fixed bounded features and statistic $S_{n,m}$ from Section 5.

(a) In the density family (6.3), let $\eta_n = v/\sqrt{n}$ for fixed $v \in \mathbb{R}^m$. Prove

$$Z_n \rightsquigarrow N_m(v, I_m), \qquad S_{n,m} \rightsquigarrow \chi^2_m(\|v\|^2),$$

and derive the limiting power of the rule $S_{n,m} > q_{m,1-\alpha}$.

(b) More generally, let the U_i be iid from $g_n(u) = 1 + n^{-1/2}r(u)$, where r is bounded, $\int_0^1 r = 0$, and n is large enough for positivity. Define $b_j = \int_0^1 \phi_j(u)r(u)\, du$, $b = (b_1, \dots, b_m)^{\mathrm{T}}$, and $\lambda_m = \sum_{j=1}^m b_j^2$. Show that

$$Z_n \rightsquigarrow N_m(b, I_m), \qquad S_{n,m} \rightsquigarrow \chi^2_m(\lambda_m). \tag{6.4}$$

Explain why λ_m is the noncentrality parameter, and prove $\lambda_m \le \int_0^1 r(u)^2\, du$.

(c) Figure 4 uses $r(u) = h\phi_1(u)$ for a scalar $h \ge 0$. Explain the ordering of the curves for $m = 1, 2, 4$. Does the figure suggest an order that is best for every departure? What calibration is needed if m is selected after seeing the data? Does a fixed-m limit justify letting m grow with n? See [8] for adaptive-order tests.

(d) For a fixed density g, use the law of large numbers to show $S_{n,m}/n \xrightarrow{\mathbb{P}} \|\mathbb{E}_g\phi(U)\|^2$. Deduce consistency when this mean vector is nonzero, and compare with Exercise 11.

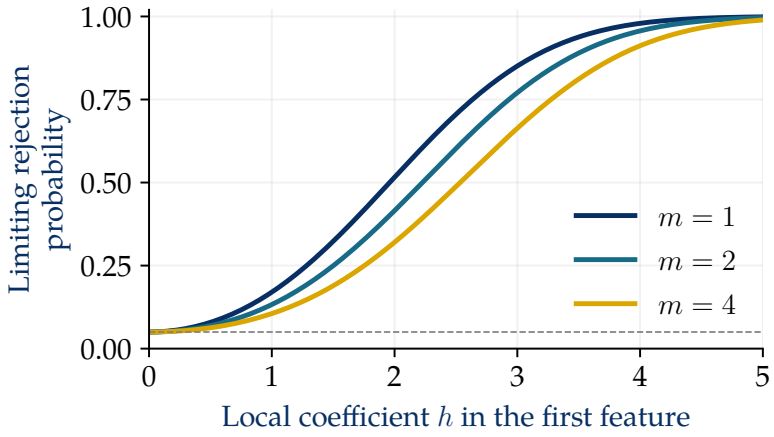


Figure 4: Exercise 9: analytic limiting power at level .05 for smooth tests with $m = 1, 2, 4$, under the local departure $r(u) = h\phi_1(u)$. Explain how the choice of order affects power. These are limiting calculations, not finite-sample simulations.

Exercise 10 (A one-feature smooth test)**.** For $\phi_1(u) = \sqrt{3}(2u - 1)$, express $S_{n,1}$ in terms of $\overline{U}$. Under the local density $g_n(u) = 1 + (a/\sqrt{n})\phi_1(u)$ for fixed scalar a, derive its limiting law and asymptotic two-sided rejection region for $\overline{U}$.

Exercise 11 (A departure outside the selected features)**.** Use normalized shifted Legendre polynomials and fix order m. Let $g(u) = 1 + \rho\phi_{m+1}(u)$ for a nonzero ρ small enough that $g > 0$. Show that the first m feature means vanish. Does this prove the test has power exactly α? Show instead that its limiting power is strictly below one at any fixed positive chi-squared cutoff. Compare with KS consistency against this fixed departure.

Exercise 12 (Grouped estimation and information)**.** For four outcomes $j = 0, 1, 2, 3$ with $p_j(\theta) = \binom{3}{j}\theta^j(1-\theta)^{3-j}$, derive the grouped MLE and the fitted Pearson degrees of freedom. For general fixed cells in a regular continuous model, explain why a full-data MLE need not give the same fitted Pearson law as a grouped MLE.

Supplementary Material

Overview of the supplement. Appendix A records the shared probability and projection tools. Appendices B–D give supporting arguments for empirical-cdf, Pearson, and smooth tests, following their order in the main text.

Guide to the proofs. The following list gives the proof location, main idea, and supporting results for each main-text result.

(i) Proposition 1.2 is proved in Section 1. The probability transform reduces a continuous simple null to uniformity.

(ii) Theorem 1.5 is proved in Section 1. We recall Week 6's empirical-cdf limits and prove the elementary concentration bound using symmetrization, a random-sign maximum, and bounded differences. Appendix B.1 proves Lemmas 1.6 and 1.7; Remark 1.8 links to proofs of sharp DKW.

(iii) Theorem 2.2 is proved in Section 2 by applying the supremum map to Week 6's process limit. Lemma B.1 proves Kolmogorov's limit-cdf formula by reflection and conditioning the Brownian-motion endpoint to zero. Appendix B also gives finite-dimensional details and local cdf drift.

(iv) Propositions 3.1, 3.3, 3.4, and 3.5 have short proofs in Section 3. They use continuous mapping, bridge reflection, and a uniform cdf expansion. The AD proof applies continuous mapping on a truncated interval and controls the omitted endpoints by their expected contributions.

(v) Theorems 4.2 and 4.3 are proved in Section 4. The multinomial CLT and Gaussian projection lemma (Appendix A) give the chi-squared limits; fitting removes the grouped-model score space. Appendix C supplies the estimator and likelihood-ratio expansions.

(vi) Theorem 5.2 is proved in Section 5 using the finite-dimensional CLT for orthonormal features. Exercises 8 and 9 ask for the score-test and local-power calculations. Appendix D gives the fitted-feature covariance argument.

Contents

A Probability and projection tools 21
B Details for empirical-cdf tests 22
C Grouped likelihood, fitted Pearson laws, and likelihood ratios 25
D Supporting results for smooth tests 26
References 27

A Probability and projection tools

These tools explain why a quadratic discrepancy has a chi-squared law. They also state exactly which convergence operation is used in the main proofs.

Lemma A.1 (Continuous mapping and small remainders; quoted). *If $Y_n \rightsquigarrow Y$ in a metric space and f is measurable and continuous at Y with probability one, then $f(Y_n) \rightsquigarrow f(Y)$. If $R_n \xrightarrow{\mathbb{P}} 0$ in a normed vector space and $Y_n \rightsquigarrow Y$, then $Y_n + R_n \rightsquigarrow Y$. For real $T_n \rightsquigarrow T$ and $c_n \xrightarrow{\mathbb{P}} c$, if $\mathbb{P}(T = c) = 0$, then $\mathbb{P}(T_n > c_n) \to \mathbb{P}(T > c)$.*

The first assertion is the continuous mapping theorem for metric spaces; see van der Vaart [1, Theorem 18.11]. The map f need not be bounded. The bounded functions in the definition of weak convergence are the test functions applied to the output $f(Y_n)$. For the finite-dimensional version and the Slutsky steps, see Week 9, Lemma G.1, p. 41. The last assertion follows by applying the first two to $T_n - c_n$, then applying the Portmanteau lemma (Week 9, Lemma 1.6(ii), p. 5) to $(0, \infty)$, whose boundary has zero probability under $T - c$. These standard results are quoted, not reproved.

Lemma A.2 (Gaussian projection). *Let Π be a real symmetric $k \times k$ orthogonal projection of rank $r \geq 1$. If $Y \sim N_k(\mu, \Pi)$ and $\Pi\mu = \mu$, then $\|Y\|^2 \sim \chi^2_r(\|\mu\|^2)$. In particular, a centered vector has squared norm χ^2_r.*

Proof of Lemma A.2. The spectral theorem gives an orthogonal matrix Q with $Q^{\mathrm{T}}\Pi Q = \mathrm{diag}(I_r, 0)$. The vector $Q^{\mathrm{T}}Y$ has its first r independent coordinates normally distributed with variance one; the remaining coordinates are constants. The condition $\Pi\mu = \mu$ makes those constants zero. The norm is invariant under Q, and the sum of squared means of the first r coordinates is $\|\mu\|^2$. □

Lemma A.3 (Covariance-standardized quadratic form). *Suppose $Y_n \rightsquigarrow N_m(0, V)$, where V is positive semidefinite of rank $r \geq 1$. Let $\widehat{V}_n \xrightarrow{\mathbb{P}} V$, and assume its Moore–Penrose inverse satisfies $\widehat{V}_n^{+} \xrightarrow{\mathbb{P}} V^{+}$; for example, this holds for consistent positive semidefinite estimates of stable rank whose nonzero eigenvalues stay away from zero. Then $Y_n^{\mathrm{T}}\widehat{V}_n^{+}Y_n \rightsquigarrow \chi^2_r$.*

Proof of Lemma A.3. Write $V = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_r, 0)Q^{\mathrm{T}}$ with $\lambda_j > 0$. For $Y \sim N_m(0, V)$, the first r coordinates of $Q^{\mathrm{T}}Y$, divided by $\sqrt{\lambda_j}$, are independent standard normals; the other coordinates are zero. Thus $Y^{\mathrm{T}}V^{+}Y \sim \chi^2_r$. Joint convergence with $\widehat{V}_n^{+}$ and continuous mapping give the conclusion. □

A raw inverse of a nearly singular estimate can amplify noise. If rank is not known, a consistent spectral threshold rule requires its own conditions. Simply claiming covariance consistency is not enough to justify an unstable inverse.

B Details for empirical-cdf tests

The main proof of the finite-sample bound is given with Theorem 1.5 in Section 1. Here we prove its two supporting lemmas. The remaining subsections explain process convergence, the bridge boundary law, and local alternatives.

B.1 Proofs of the concentration lemmas

The random-sign maximum controls the ordered threshold process; bounded differences turns the mean bound into a tail bound. Lemmas 1.6 and 1.7 are stated in the proof of Theorem 1.5; their proofs follow here.

Proof of Lemma 1.6. Fix $a > 0$, and let A_k be the event that k is the first index with $|S_k| \geq a$. The events A_k are disjoint. Independence and the zero means of the remaining signs give

$$\mathbb{E}\{S_n^2 \mid \epsilon_1, \ldots, \epsilon_k\} = S_k^2 + n - k.$$

Consequently,

$$n = \mathbb{E}S_n^2 \geq \sum_{k=1}^{n} \mathbb{E}\{S_n^2 \mathbf{1}_{A_k}\} \geq a^2 \mathbb{P}\left\{ \max_{0 \leq k \leq n} |S_k| \geq a \right\}.$$

Writing $M = \max_k |S_k|$ and integrating this tail bound yields

$$\mathbb{E}M = \int_0^\infty \mathbb{P}(M > t)\, dt \leq \sqrt{n} + \int_{\sqrt{n}}^\infty \frac{n}{t^2}\, dt = 2\sqrt{n}. \qquad \square$$

Proof of Lemma 1.7. First let Y have mean zero and take values in an interval of length c. For $h(\lambda) = \log \mathbb{E}e^{\lambda Y}$, we have $h(0) = h'(0) = 0$. Its second derivative is the variance of Y under the probability law with weights proportional to $e^{\lambda Y}$. This law has the same range, so its variance is at most $c^2/4$: subtract the interval midpoint and bound the resulting squared deviation by $c^2/4$. Integrating twice gives

$$\mathbb{E}e^{\lambda Y} \leq e^{\lambda^2 c^2/8}, \qquad \lambda \geq 0.$$

Set $M_i = \mathbb{E}(Z \mid X_1, \ldots, X_i)$, $M_0 = \mathbb{E}Z$, and $D_i = M_i - M_{i-1}$. Conditional on $X_1, \ldots, X_{i-1}$, the increment D_i has mean zero and lies in an interval of length at most c_i. Indeed, independence allows the unseen coordinates to be integrated out using the same law for any two values of X_i, so the conditional expectation changes by at most c_i. Apply the preceding exponential bound conditionally and iterate to obtain

$$\mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \exp\left(\frac{\lambda^2}{8} \sum_i c_i^2 \right).$$

Markov's inequality bounds the tail by $\exp\{-\lambda t + \lambda^2 \sum_i c_i^2/8\}$. Taking $\lambda = 4t/\sum_i c_i^2$ proves the claim. $\square$

B.2 Finite-dimensional convergence and what it does not prove

For fixed $u_1, \ldots, u_q \in [0,1]$, define the iid bounded vector

$$Y_i = (\mathbf{1}\{U_i \leq u_1\} - u_1, \ldots, \mathbf{1}\{U_i \leq u_q\} - u_q)^{\mathrm{T}}.$$

Under uniformity, its mean is zero and its covariance has entries $\min(u_a, u_b) - u_a u_b$. The multivariate CLT gives $(\alpha_n(u_1), \ldots, \alpha_n(u_q)) \rightsquigarrow (B(u_1), \ldots, B(u_q))$. This identifies every finite-dimensional limiting law.

It does not by itself imply convergence of $\sup |\alpha_n|$. That supremum ranges over infinitely many thresholds. The empirical-process theorem adds asymptotic control of the oscillations, often expressed as asymptotic equicontinuity, together with tightness in the supremum-norm space. Those are the additional ingredients in Donsker's theorem recalled in Theorem 1.5. We use the theorem with its hypotheses rather than treating the finite-dimensional calculation as a full proof.

B.3 The Brownian-bridge boundary law

This classical result supplies the distribution formula used after the process-convergence argument in Theorem 2.2.

Lemma B.1 (Kolmogorov's Brownian-bridge formula)**.** *Let B be a standard Brownian bridge on $[0,1]$ and set $K = \sup_{0\le u\le 1}|B(u)|$. Then, for every $c>0$,*

$$\mathbb{P}(K \le c) = 1 - 2\sum_{j=1}^{\infty}(-1)^{j-1}e^{-2j^2c^2}.$$

The formula is due to Kolmogorov [6]. For a modern derivation, see Perman and Wellner (2014), Section 3.1, p. 3112 [9]. Their unnumbered formulas following (3.3) give the cdf and upper tail of the absolute bridge maximum.

Proof of Lemma B.1. Let W be standard Brownian motion, let φ be the standard normal density, and put $\tau = \inf\{t \ge 0 : |W(t)| = c\}$. We first compute the probability that W stays in $(-c,c)$ until time one and ends near zero.

For $0<\varepsilon<c$, define the bounded function

$$h_\varepsilon(z) = \sum_{j\in\mathbb{Z}}(-1)^j \mathbf{1}\{|z-2jc|<\varepsilon\}.$$

The intervals are disjoint, so at most one summand is nonzero. Inside $(-c,c)$, this function equals $\mathbf{1}\{|z|<\varepsilon\}$. Reindexing the sum shows that it is odd about either boundary:

$$h_\varepsilon(a+z) = -h_\varepsilon(a-z), \qquad a\in\{-c,c\}.$$

After the first boundary hit, future Brownian increments are independent of the past and symmetric about zero; this is the strong Markov property. Reflecting those increments therefore preserves the path law and changes the sign of $h_\varepsilon(W(1))$. Thus paths with $\tau\le 1$ contribute zero to its expectation. On $\{\tau>1\}$, the endpoint lies inside $(-c,c)$, giving

$$\mathbb{P}\{\tau>1,\ |W(1)|<\varepsilon\} = \mathbb{E}h_\varepsilon(W(1)) = \sum_{j\in\mathbb{Z}}(-1)^j\int_{2jc-\varepsilon}^{2jc+\varepsilon}\varphi(y)\,dy.$$

Divide by $\mathbb{P}\{|W(1)|<\varepsilon\}$ and let $\varepsilon\downarrow 0$. Gaussian decay gives a summable bound uniform for $\varepsilon<c/2$, so the limit passes through the sum:

$$\lim_{\varepsilon\downarrow 0}\mathbb{P}\{\tau>1 \mid |W(1)|<\varepsilon\} = \sum_{j\in\mathbb{Z}}(-1)^j\frac{\varphi(2jc)}{\varphi(0)} = \sum_{j\in\mathbb{Z}}(-1)^j e^{-2j^2c^2}.$$

To identify the left side, write $B(t) = W(t) - tW(1)$. This bridge is independent of $W(1)$, and on $\{|W(1)|<\varepsilon\}$ its supremum distance from W is less than ε. Hence

$$\mathbb{P}(K\le c-\varepsilon) \le \mathbb{P}\{\tau>1 \mid |W(1)|<\varepsilon\} \le \mathbb{P}(K<c+\varepsilon).$$

At every continuity point c of the cdf of K, the limit is therefore $\mathbb{P}(K\le c)$. The Gaussian series is locally uniformly convergent for $c>0$, hence continuous. Since cdf continuity points are dense and every cdf is right-continuous, the identity extends to every $c>0$. Pairing the terms j and $-j$ gives the stated formula. □

B.4 Local cdf drift

To compare cdf tests with the local chi-squared limits, consider iid U_i with cdf

$$G_n(u) = u + n^{-1/2}H(u), \quad H(u) = \int_0^u h(v)\,dv,$$

where h is bounded and $\int_0^1 h = 0$. For large n, G_n is a cdf with positive density. Using the probability transform within each row and $\|G_n - \mathrm{id}\|_\infty \to 0$, Donsker's theorem and stochastic equicontinuity give

$$\sqrt{n}\{\widehat{G}_n(u) - u\} \rightsquigarrow B(u) + H(u).$$

Here $\widehat{G}_n$ is the empirical cdf of the row-n sample; G_n is its population cdf. Consequently, the local KS limit is $\sup_u |B(u) + H(u)|$, while the local CvM limit is $\int_0^1 \{B(u) + H(u)\}^2\,du$. Unlike a finite-dimensional smooth test, KS sees the integrated density departure H over the whole interval. This explains why the geometry of a departure, and not just its density L^2 norm, matters for comparisons.

C Grouped likelihood, fitted Pearson laws, and likelihood ratios

This appendix first derives the grouped estimator expansion needed in Theorem 4.3, then compares it with full-data estimation. All cells remain fixed and k, d are fixed.

C.1 Supporting estimator expansion

Proposition C.1 (Grouped multinomial MLE expansion)**.** *Let $p(\theta)$ be twice continuously differentiable near an interior θ_0, with all probabilities positive, full-rank derivative J, and $I_g = J^{\mathrm{T}} D^{-1} J$ nonsingular. Suppose a consistent interior grouped MLE exists and its normalized Hessian converges to $-I_g$ uniformly between θ_0 and $\widehat{\theta}$. Then it satisfies* (4.2).

Proof of Proposition C.1. Omit the multinomial coefficient and write the log likelihood as

$$\ell_g(\theta) = \sum_j N_j \log p_j(\theta).$$

Since $\mathbf{1}^{\mathrm{T}} J = 0$, its normalized score at θ_0 is

$$n^{-1/2}\dot{\ell}_g(\theta_0) = J^{\mathrm{T}} D^{-1} \sqrt{n}(\widehat{p} - p) = A^{\mathrm{T}} Z_n.$$

The score equation and its integral Taylor expansion give

$$0 = n^{-1/2}\dot{\ell}_g(\theta_0) + \left\{ \int_0^1 n^{-1}\ddot{\ell}_g(\theta_0 + t(\widehat{\theta} - \theta_0))\,dt \right\} \sqrt{n}(\widehat{\theta} - \theta_0).$$

The matrix in braces tends to $-I_g$ and is invertible with probability tending to one. Solving and using Slutsky gives $\sqrt{n}(\widehat{\theta} - \theta_0) = I_g^{-1} A^{\mathrm{T}} Z_n + o_{\mathbb{P}}(1)$, which is (4.2). □

For a consistent regular minimum-Pearson-distance estimator, the first-order condition yields $A^{\mathrm{T}}\{Z_n - A\sqrt{n}(\widehat{\theta} - \theta_0)\} = o_{\mathbb{P}}(1)$. The same expansion follows. Conditions that guarantee existence and consistency must still be checked for the chosen model.

C.2 Why a full-data MLE differs

Suppose the observations originally have a regular density f_θ, score $s(X)$ and full-data information $I_f = \mathbb{E}(ss^{\mathrm{T}})$. Its efficient MLE has influence function $I_f^{-1}s$. The standardized cell-indicator vector for one observation is

$$Y(X) = D^{-1/2}\{(\mathbf{1}\{X \in C_j\})_{j=1}^k - p\}.$$

Differentiation under the integral gives $\mathbb{E}[Ys^{\mathrm{T}}] = A$. Hence the fitted standardized residual is asymptotically an average of $Y - AI_f^{-1}s$ and has covariance

$$\Sigma_f = I_k - tt^{\mathrm{T}} - AI_f^{-1}A^{\mathrm{T}}. \tag{C.1}$$

The grouped information $I_g = A^{\mathrm{T}}A$ is at most the full-data information in the positive-semidefinite order. One way to see this is that the grouped score is $\mathbb{E}[s(X) \mid C(X)]$, where $C(X)$ is the cell label, and the law of total covariance gives $I_f - I_g = \mathbb{E}\,\mathrm{Var}(s \mid C) \succeq 0$.

If $I_f \neq I_g$, the matrix in (C.1) is generally not a projection. Its Gaussian squared norm is a weighted sum $\sum_l \nu_l Z_l^2$ over its positive eigenvalues ν_l, not necessarily χ^2_{k-1-d}. The full-data estimate can use within-cell information which the grouped vector does not contain. This is the precise reason for specifying the estimation method in the fitted theorem.

C.3 Technical expansion for the multinomial likelihood ratio

Proposition C.2 (Pearson and multinomial likelihood ratio)**.** *Under the fixed positive simple multinomial null, let*

$$G_n^2 = 2\sum_{j=1}^k N_j \log\frac{N_j}{np_j}, \qquad 0\log 0 := 0.$$

Then $G_n^2 = X_n^2 + o_{\mathbb{P}}(1)$ and $G_n^2 \rightsquigarrow \chi^2_{k-1}$. For a regular grouped fit as in Theorem 4.3, replacing p by $p(\widehat{\theta})$ yields $G^2_{n,\mathrm{fit}} = X^2_{n,\mathrm{fit}} + o_{\mathbb{P}}(1)$.

Proof of Proposition C.2. Let $\delta_j = (\widehat{p}_j - p_j)/p_j$. Since k is fixed and $p_j > 0$, $\max_j |\delta_j| = O_{\mathbb{P}}(n^{-1/2})$. With probability tending to one it is at most $1/2$, and a uniform Taylor expansion gives

$$(1+\delta)\log(1+\delta) = \delta + \tfrac{1}{2}\delta^2 + O(|\delta|^3).$$

Thus $G_n^2 = 2n\sum_j p_j\delta_j + n\sum_j p_j\delta_j^2 + O_{\mathbb{P}}(n^{-1/2})$. The linear term is zero because the probabilities sum to one. The quadratic term is X_n^2. For the fitted case the fitted probabilities stay bounded away from zero with probability tending to one, the residuals are root-n by the fitted expansion, and their sum is also zero. The identical Taylor argument applies. □

This is the goodness-of-fit version of the first-order equivalence of likelihood and quadratic tests in Week 9. It is an asymptotic statement and does not make the finite-sample statistics equal.

D Supporting results for smooth tests

D.1 Fitted features and nuisance scores

For this fitted extension, let f_θ be a regular density model with continuous cdf F_θ. Define

$$q_\theta(x) = \phi(F_\theta(x)) = (\phi_1(F_\theta(x)), \ldots, \phi_m(F_\theta(x)))^{\mathrm{T}}.$$

This is the vector of chosen features evaluated after transforming x by the model cdf. Write $\mathbb{E}_\theta$ for expectation when $X \sim F_\theta$. Since $F_\theta(X)$ is then uniform, the zero-mean condition in (5.1) gives

$$\mathbb{E}_\theta q_\theta(X) = \int_0^1 \phi(u)\, du = 0.$$

Both the transform and the sampling law use the same parameter θ. At the true parameter θ_0, write $s(x) = \nabla_\theta \log f_\theta(x)|_{\theta=\theta_0}$ for the model score, $I = \mathbb{E}_{\theta_0}(ss^{\mathrm{T}})$ for its nonsingular information matrix, and $C = \mathbb{E}_{\theta_0}(q_{\theta_0} s^{\mathrm{T}})$. We justify the expansion

$$\frac{1}{\sqrt{n}} \sum_i q_{\widehat{\theta}}(X_i) = \frac{1}{\sqrt{n}} \sum_i \{q_{\theta_0}(X_i) - CI^{-1} s(X_i)\} + o_{\mathbb{P}}(1). \tag{D.1}$$

Proposition D.1 (Efficient plug-in smooth covariance)**.** *Assume differentiability under expectation, a uniform law for the derivative of q_θ near θ_0, a Taylor remainder $o_{\mathbb{P}}(1)$ for the normalized sum after substituting $\widehat{\theta}$, and the efficient MLE expansion. Then (D.1) holds and its limiting covariance is $V = I_m - CI^{-1}C^{\mathrm{T}}$. If $r = \operatorname{rank} V \geq 1$ and a covariance estimate satisfies the conditions of Lemma A.3, the covariance-standardized statistic converges to χ^2_r.*

Proof of Proposition D.1. Write $\dot{q}_\theta(x) = \partial q_\theta(x)/\partial \theta^{\mathrm{T}}$ and $s_\theta(x) = \nabla_\theta \log f_\theta(x)$. In the zero-mean identity, both q_θ and the density vary with θ. Differentiating their product under the integral gives

$$\begin{aligned} 0 &= \frac{\partial}{\partial \theta^{\mathrm{T}}} \int q_\theta(x) f_\theta(x)\, dx \\ &= \int \{\dot{q}_\theta(x) f_\theta(x) + q_\theta(x) f_\theta(x) s_\theta(x)^{\mathrm{T}}\}\, dx \\ &= \mathbb{E}_\theta \dot{q}_\theta(X) + \mathbb{E}_\theta\{q_\theta(X) s_\theta(X)^{\mathrm{T}}\}. \end{aligned}$$

At $\theta = \theta_0$, the second term is C, so $\mathbb{E}_{\theta_0} \dot{q}_{\theta_0}(X) = -C$. The empirical derivative law and Taylor remainder therefore give

$$n^{-1/2} \sum_i q_{\widehat{\theta}}(X_i) = n^{-1/2} \sum_i q_{\theta_0}(X_i) - C\sqrt{n}(\widehat{\theta} - \theta_0) + o_{\mathbb{P}}(1).$$

Substitute the MLE expansion. Since $\operatorname{Var}(q_{\theta_0}) = I_m$, $\mathbb{E}(q_{\theta_0} s^{\mathrm{T}}) = C$, and $\operatorname{Var}(s) = I$, expanding the covariance gives $I_m - CI^{-1}C^{\mathrm{T}}$. The iid multivariate CLT and Lemma A.3 give the last claim. □

The residual feature $q - CI^{-1}s$ is orthogonal in $L^2(F_{\theta_0})$ to every nuisance-score coordinate, since its covariance with s is $C - CI^{-1}I = 0$. If a feature lies exactly in the nuisance-score span, it has no residual variance. Partial overlap can instead reduce an eigenvalue without making it zero. This distinguishes covariance correction from blindly subtracting a parameter count.

References

[1] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998.

[2] Jerzy Neyman. "Smooth Test" for goodness of fit. *Skandinavisk Aktuarietidskrift*, 20(3–4):149–199, 1937.

[3] J. C. W. Rayner, Olivier Thas, and D. J. Best. *Smooth Tests of Goodness of Fit: Using R*. Wiley, 2 edition, 2009.

[4] Pascal Massart. The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality. *The Annals of Probability*, 18(3):1269–1283, 1990.

[5] Henry W. J. Reeve. A short proof of the Dvoretzky–Kiefer–Wolfowitz–Massart inequality. https://arxiv.org/abs/2403.16651, 2024. See Sections 1–2 and Corollary 2.

[6] A. N. Kolmogorov. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell'Istituto Italiano degli Attuari*, 4:83–91, 1933.

[7] T. W. Anderson and D. A. Darling. Asymptotic theory of certain "Goodness of Fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics*, 23(2):193–212, 1952.

[8] W. C. M. Kallenberg and Teresa Ledwina. On data driven Neyman's tests. *Probability and Mathematical Statistics*, 15:409–426, 1995.

[9] Mihael Perman and Jon A. Wellner. An excursion approach to maxima of the Brownian bridge. *Stochastic Processes and their Applications*, 124(9):3106–3120, 2014.